DPP4SiPM command line documentation

The interaction with the DPP4SiPM multichannel analyzer (MCA) is performed through ASCII commands. Either through Ethernet or UART, the same command line applies to setup the device parameters or to retrieve data. The commands requests sent from the host (computer or device that controls the MCA) are always replied by the MCA upon a successful communication establishment.

1. ASCII commands format

All the commands share the following structure, as described next.

1.1 Request (from host to MCA)

<Start character> <Command Code> <Parameters and values > <End character>

<Start character>: Always the dollar sign: `$` (ASCII 0x24)
<Command code>: Two-letter abbreviation
<Parameters and values>: 32-bit values separated by space character (ASCII 0x20)
<End character>: Carriage return `\n` (ASCII 0x0D)

Example:
```
$SP 8 52921 3 \r
```

- the command `$SP` means: *set parameter*
- the `8` means: *parameter #8* (see the section MCA commands for details)
- the `52921` means: -0.7699584 in signed Q2.14 fixed-point representation
- the `3` means: binary flag (0b011)
- the `\r` means: end character, equivalent to ASCII code 0x0D

1.2 Reply (from MCA to host)

The MCA with a similar format, starting with a distinctive reply start character
<Start character> <Received request command> <Values> <End characters>

<Start character>: Always the exclamation mark: `!` (ASCII 0x21)
<Received request command>: The same two-character command from the request
<Values>: The payload of the reply. Contents depend on the request command sent.
<End characters>: Two successive characters are returned always: `\n\r` (ASCII 0x0D 0x0A)

Example:
If the request is `$SP 8 52921 3 \r`, the reply might be:
```
$SP\n\r
```

For the request `$RT -1`, the reply might be:
```
!RT 0 0 0 0 10000000 176\n\r
```

1.3 Error replies

In case the request command is wrong or has not been delivered correctly, an error reply will be sent from the MCA back to the host.

The error reply format is defined as follows:
<Start character> ERROR:<Error code><End characters>

<Start character>: Always the exclamation mark: **!** (ASCII 0x21)
ERROR: text followed by the error code
<Error code>: **0** if the request command is not recognized, **1** if the sent parameters are wrong
<End characters>: **\n\r** (ASCII 0x0D 0x0A)

Example:
If the request is `$KR \r`, the reply should be:
```
$ERROR:00\n\r
```
because *KR* is an unknown request command

If the request is `$SP 8 52921 \r`, *t*he reply might be:
```
$ERROR:01\n\r
```
because one value is missing to set the parameter #8 (two expected)

2 MCA commands

The MCA is configured through request commands. In the same way, data such as the spectrum and timer values can be retrieved using the command line. Such request commands always follow the aforementioned format, and are next.

2.1 Get Version ($GV)

Returns the firmware version of the system.

2.1.1 Parameters

No parameters are required

2.1.2 Example

```
$GV\r
!GV 1.0\n\r
```

2.2 Get Serial Number ($SN)

Returns the serial number of the MCA.

2.2.1 Parameters

No parameters are required

2.2.2 Example

```
$SN\r
!SN 210328B00000B\n\r
```

2.3 Get Parameters ($GP)

Returns the values from a specific DPP parameters group in the MCA. Refer to the section Digital pulse processor for details on the available options.

2.3.1 Parameters

Only one parameter is accepted: the DPP parameter group index to be read. The available parameter groups are indexed as follows:

1. Error: Reference source not found
2. Peak detector – slow
3. Scope
4. Timers
5. Baseline restorer - slow
6. Scope mux
7. DCS slow (deprecated)
8. Formatter (preprocessing)
9. Pulse shaper – fast
10. Baseline restorer - fast
11. DCS – fast (deprecated)
12. Peak detector – fast
13. Pileup rejector
14. High Voltage (only for PMT version)
15. Variable-gain analog amplifier (VGA)

2.3.2 Example

For instance, to retrieve the status of the Formatter (preprocessing) block, the parameter group #8 should be called:

```
$GP 8\r
```

The expected reply should be something like this:

```
!GP 52921 0\n\r
```

2.4 *Read* Timers ($RT)

Reads the current status/value of one or all the timers in the MCA.

2.4.1 Parameters

The timer index (address) to be read.

Special case: to read all the timers simultaneously use -1. The reply will include the following fields:

Timer A, Timer B, Timer C, Status Register, Preset (collection time), Control bits (flags)

2.4.2 Example

```
$RT -1\r
!RT 60000 60005 60000 4 300000 244\n\r
```

2.5 Read Spectrum ($RM)

Reads the BRAM memory values where the spectrum is stored. The reply command contains the !R\n\r header, followed by exactly 8 KB of binary data. The binary data represents the 2048 channels of the spectrum in unsigned 32-bit representation (little endian coding).

2.5.1 Parameters

The base address of the memory to read the spectrum from. The current version supports only 0.

2.5.2 Example

```
$RM 0\r
!R\n\r [8 KB of binary data: 32-bit words in little endian]
```

2.6 Load Scope ($LS)

Reads the two channels (signed 16-bit each) of the oscilloscope (CH1 and CH2). The data is replied from the MCA in a 32-bit binary (little endian) data stream. CH1 is stored in the first 16 bits, while CH2 can be retrieved by reading the most significant 16 bits. The length of the serial stream is determined by the BRAM Size parameter (see the subsection Scope), by default 2048 samples.

With the default configuration, 8 KB (2048 samples for the 2 channels in 32-bit words) is streamed from the MCA to the host.

2.6.1 Parameters

No parameters are required.

2.6.2 Example

```
$LS\r
!L\n\r [8 KB of binary data: 32-bit words in little endian]
```

2.7 Set Parameters ($SP)

Sets the values for a group of the DPP parameters in the MCA. Refer to the section Digital pulse processor for details on the available options.

2.7.1 Parameters

The first parameter is the DPP parameter group index, followed by the values of the corresponding parameter group. The DPP parameter group index is shown next:

1. Error: Reference source not found
2. Peak detector – slow
3. Scope
4. Timers
5. Baseline restorer - slow
6. Scope mux
7. DCS slow (deprecated)
8. Formatter (preprocessing)
9. Pulse shaper – fast
10. Baseline restorer - fast
11. DCS – fast (deprecated)
12. Peak detector – fast
13. Pileup rejector
14. High Voltage (only for PMT version)
15. Variable-gain analog amplifier (VGA)

2.7.2 Example

For instance, to set the values for the Pulse Shaper Slow (DPP parameter group 1):

```
$SP 1 20 220 1145 2500 1000 33554432 32973417 2097 122 172 16443768 0 0 15319287 0 0 0\n\r
```

The expected reply should be something like this:

```
!SP
```

Another example involves a parameter group with less fields/values: the Pileup Rejector (DPP parameter group 13):

```
$SP 13 237 1 20\n\r
```

The expected reply should be something like this:

```
!SP
```

2.8 Acquisition Start/Stop ($AQ)

Starts/stop the spectrum acquisition. It also controls the timers' reset values.

2.8.1 Parameters

One register is required to be sent as a parameter: the setting flag. This flag value must be chosen out of one of the following values.

Flag value	**Description**
0	Starts automatic acquisition. Cleans BRAM contents prior to starting. Timers leverage ping-pong spectrum memory access for precise high count-rate applications.
1	Starts acquisition manually. Cleans BRAM contents prior to starting. Only one BRAM memory segment is used to store the spectrum. Used in most of the measurement situations.
2	Stops spectrum acquisition manually.
4	Clears all the timer counters. Starts from 0.

2.8.2 Example

```
$AQ 4\r
!AQ\n\r
```

2.9 Clear Spectrum ($CS)

Clears the BRAM memory segment contents where the spectrum is stored.

2.9.1 Parameters

Memory segment index to clear. The current version only supports memory segment 0, no matter which parameter (still required) value is sent.

2.9.2 Example

```
$CS 0\r
!CS\n\r
```

2.10 Set Serial Number ($SS)

Not implemented yet.

2.11 Keep Alive (~~~)

A special signaling packet to periodically check whether the MCA is available/alive. Mainly used in threaded applications or through graphical user interfaces. No parameters are used.

2.11.1 Example

```
~~~\r
~~~\n
```

3 Digital pulse processor

The digital pulse processor (DPP) in the MCA is configurable, aiming at matching the detector characteristics for each use case. The configurable parameters can be retrieved or set by using the request commands `$GP` and `$SP`, respectively. These values are grouped according to the internal building blocks of the DPP and should be appended sequentially in **unsigned 32-bit format** every time the `$SP` command is sent.

3.1 Pulse shaper – slow

Parameter group: 1

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
Tclk	Period of the DPP clock in the FPGA, **in nanoseconds** (ns).	U – Q32.0	20	20
Taur	Rising edge constant of the detector pulse shape, **in ns.**	U – Q32.0	220	220
Taud	Falling edge (decay time) constant of the detector pulse shape, **in ns.**	U – Q32.0	1145	1145
Taupk	Peaking time of the trapezoidal shaper, **in ns.**	U – Q32.0	3000	3000
Taupk_top	Flat-top of the trapezoidal filter, **in ns.**	U – Q32.0	0	0
Gain	Digital gain at the output of the trapezoidal filter.	U – Q1.25	33554432	1.0
b10	exp(-Tclk/Taud)	U – Q0.25	32973417	0.9826844036579132
na_inv	Tclk/taupk	U – Q0.18	1747	0.006664276123046875
na	Taupk/Tclk	U – Q10.0	147	147
nb	(Taupk+Taupk_top)/Tclk	U – Q10.0	147	147
b20	2-pole normalization	U – Q4.21	16200338	7.724923133850098
~~dc_offset_1~~	***Deprecated.*** DC offset. Now applied in Formatter (Parameter group 8)	S – Q2.14	0	0
b2	3-pole compensation for PMT detectors	U – Q0.25	0	0
b1	Sum of exponentials of 2- and 3-pole decays	U – Q1.24	16486709	0.9826844334602356

aa20	Sum of exponentials for 3-pole compensation in PMT	U – Q0.25	0	0
flags	Invert input signal (bit 0)	U – Q32.0	0	0
~~dc_offset_2~~	***Deprecated.*** DC offset after shaper.	S – Q11.14	0	0

3.2 Peak detector – slow

Parameter group: 2

Setting	**Description**	**Signedness and fixed-point format (Qm.n)**	**Default value (unsigned 32-bit)**	**Default value (human readable)**
Tclk	Period of the DPP clock in the FPGA, in **nanoseconds** (**ns**).	U – Q32.0	20	20
Blanking_time	Period to disable peak detector after an event detection, expressed in **clock cycles (Tclk units)**. Requires b*lanking time factor* to be computed.	U – Q32.0	135	135
Time_over_threshold	ToT for peak detector, in **milliseconds**. Requires Time Over Threshold factor. (tau_pk_top_fast + tot_factor*tau_pk_fast)/Tclk	U – Q2.14	65	0.00396728515625
x_min	Low-level discriminator (LLD), in **Volts**.	S – Q2.14	245	0.01495361328125
x_max	High-level discriminator (HLD), in **Volts**.	S – Q2.14	32061	1.95684814453125
flags	Enable peak detector subsystem (bit 0)	U – Q32.0	1	1

3.3 Scope

Parameter group: 3

Setting	**Description**	**Signedness and fixed-point format (Qm.n)**	**Default value (unsigned 32-bit)**	**Default value (human readable)**
Tclk	Period of the DPP clock in the FPGA, in **nanoseconds** (**ns**).	U – Q32.0	20	20
BRAM_Size	32-bit words to be stored (scope trace length)	U – Q32.0	2048	2048
Threshold	Threshold level of the scope, in **Volts**.	S – Q2.14	655	0.03997802734375
Delay	Trigger delay, in **clock cycles (Tclk units**)	U – Q32.0	1000	1000
Enable	Enable oscilloscope flag	U – Q32.0	1	1
Clear	Clear initial trace	U – 32.0	1	1
~~Full~~	**DEPRECATED**. Status of the FIFO.	U – 32.0	1	1
Down-sample	Bit masks - 0x00=Tclk/1 - 0x01=Tclk/2 - 0x02=Tclk/3 - 0x03=Tclk/4 Bit offsets - Down-sampling=0x00 - Enable flag=0x02	U – 32.0	4	4 (no down-sampling, but enabled)
Sampling_mode	Bit masks - 0x00=decimate - 0x01=take max sample value between 2 clocks - 0x02=take minimal sample between two clocks Bit offsets - CH1=0x00 - CH2=0x02 - TRIG = 0x04	U – 32.0	21	0x01 << 0x00 + 0x01 << 0x02 + 0x01 << 0x04 = 21 (CH1, CH2, and trigger): take max. sample value between 2 clocks
Trigger_mode	Normal mode only = 0	U – 32.0	0	0

3.4 Timers

Parameter group: 4

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
Preset	Collection time, in **ms**. Measured using **Timer C**.	U – Q32.0	10000000	10000000
Ctrl_bits LT = live time RT = real time	<u>Bit offsets</u> - 0x00=manual mode - 0x01=auto mode - 0x02=TMR C: 1=LT, 0=RT - 0x03=TMR B: 1=LT, 0=RT - 0x04=TMR A: 1=LT, 0=RT - 0x05=TMR C enable - 0x06=TMR B enable - 0x07=TMR A enable - 0x08=TMR C clear - 0x09=TMR B clear - 0x0A=TMR A clear	U – Q32.0	176	Manual mode, Timer A for LT and Timer C for RT. All the timers initially disabled.

3.5 Baseline restorer - slow

Parameter group: 5

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
Thresholds (high | low)	The 32-bit word is split in 16-bit words for each threshold (in **Volts**): (High Thrshld | Low Thrshld)	S – Q2.14 S – Q2.14	64717	High threshold = 0.00 Low threshold = -0.04998779296875
Flags	BLR control. The speed bitmasks are estimated with a DPP clock of 50 MHz. <u>BLR speed bitmask</u> 0x00=Tau BLR ~ 1.28 μs ($1/2^6$) 0x01=Tau BLR ~ 10.2 μs ($1/2^9$) 0x02=Tau BLR ~ 81.9 μs ($1/2^{12}$) 0x03=Tau BLR ~ 1.31 ms ($1/2^{16}$) <u>Bit offsets</u> 0x00=BLR reset	U – Q32.0	10	- BLR not in reset state - BLR speed ~ 10.2 μs - BLR enabled

	0x01=BLR speed 0x03=BLR enable			
Threshold_gain	BLR gain, used to fine-tune the sensitivity of BLR.	U – Q8.8	768	3.0
Preset	Look-ahead preset. Split in 3 registers: see bit offsets. All values are expressed in clock cycles (**Tclk units**) <u>Bit offsets</u> - 0x00=Slow shaper time (2*tau_pk_slow+tau_pk_top_slow) - 0x0A=width limit (3*slow shaper time) - 0x14=trailing edge (shaper_slow_taupk / 2)	U – Q10.0 U – Q10.0 U – Q10.0	1970476	- Slow shaper time = 300 - Width limit = 900 - Trailing edge = 1
b0	BLR charge time (in **milliseconds**)	U – Q0.32	8589	1.99978239834308e-06
a1	BLR discharge time (in **milliseconds**)	U – Q0.32	4294967123	0.9999999597202986
threshold_gain_low	Fine gain for BLR gain.	U – Q8.8	12800	50

3.6 Scope mux

Parameter group: 6

Setting	**Description**	**Signedness and fixed-point format (Qm.n)**	**Default value (unsigned 32-bit)**	**Default value (human readable)**
Scope Mux	Multiplexer for scope input channels. 4 bits for each channel: (CH2 | CH1) <u>Bit masks</u> Check the latest Vivado design to determine which inputs are connected to each of the 16 channels. <u>Bit offsets</u> 0x00=CH1 input select 0x04=CH2 input select	U – Q4.0 U – Q4.0	19	CH1 = Input index 3 CH2 = Input index 1

3.7 DCS slow (deprecated)

Parameter group: 7

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
~~Threshold_high~~	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**
~~Htreshold_low~~	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**
~~Enable_shift~~	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**

3.8 Formatter (preprocessing)

Parameter group: 8

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
DC_Offset	DC offset applied to ADC input, expressed in **Volts**	S – Q2.14	52921	-0.76995849609375
Flags	Pulse polarity inversion and moving average window length (smoothing) selection. <u>Bit masks for smoothing</u> 0x00=Smoothing x1 (no filtering) 0x01=Smoothing x2 0x02=Smoothing x4 0x03=Smoothing x8 <u>Bit masks for pulse inversion</u> 0x00=Invert pulse <u>Bit offsets</u> 0x00=Pulse inversion 0x01=Smoothing	U – Q32.0	4	- Inversion = No - Smoothing = x4

3.9 Pulse shaper – fast

Parameter group: 9

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
Tclk	Period of the DPP clock in the FPGA, **in nanoseconds** (ns).	U – Q32.0	20	20
Taur	Rising edge constant of the detector pulse shape, **in ns.**	U – Q32.0	220	220
Taud	Falling edge (decay time) constant of the detector pulse shape, **in ns.**	U – Q32.0	1145	1145
Taupk	Peaking time of the trapezoidal shaper, **in ns.**	U – Q32.0	200	200
Taupk_top	Flat-top of the trapezoidal filter, **in ns.**	U – Q32.0	0	0
Gain	Digital gain at the output of the trapezoidal filter.	U – Q1.25	33554432	1.0
b10	exp(-Tclk/Taud)	U – Q0.25	32973417	0.9826844036579132
na_inv	Tclk/taupk	U – Q0.18	26214	0.09999847412109375
na	Taupk/Tclk	U – Q10.0	7	7
nb	(Taupk+Taupk_top)/Tclk	U – Q10.0	7	7
b20	2-pole normalization	U – Q4.21	16443768	7.840999603271484
~~dc_offset_1~~	***Deprecated.*** DC offset. Now applied in Formatter (Parameter group 8)	S – Q2.14	0	0
b2	3-pole compensation for PMT detectors	U – Q0.25	0	0
b1	Sum of exponentials of 2- and 3-pole decays	U – Q1.24	15319287	0.9131006598472595
aa20	Sum of exponentials for 3-pole compensation in PMT	U – Q0.25	0	0
flags	Invert input signal (bit 0)	U – Q32.0	0	0
~~dc_offset_2~~	***Deprecated.*** DC offset after shaper.	S – Q11.14	0	0

3.10 Baseline restorer - fast

Parameter group: 10

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
Thresholds (high | low)	The 32-bit word is split in 16-bit words for each threshold (in **Volt units**): (High Thrshld | Low Thrshld)	S – Q2.14 S – Q2.14	64717	High threshold = 0.00 Low threshold = -0.04998779296875
Flags	BLR control. The speed bitmasks are estimated with a DPP clock of 50 MHz. <u>BLR speed bitmask</u> 0x00=Tau BLR ~ 1.28 µs ($1/2^6$) 0x01=Tau BLR ~ 10.2 µs ($1/2^9$) 0x02=Tau BLR ~ 81.9 µs ($1/2^{12}$) 0x03=Tau BLR ~ 1.31 ms ($1/2^{16}$) <u>Bit offsets</u> 0x00=BLR reset 0x01=BLR speed 0x03=BLR enable	U – Q32.0	0	- BLR not in reset state - BLR speed ~ 1.28 µs - BLR disabled
Threshold_gain	BLR gain, used to fine-tune the sensitivity of BLR.	U – Q8.8	1536	6.0
b0	BLR charge time (in **milliseconds**)	U – Q0.32	85899	1.999991945922375e-05
a1	BLR discharge time (in **milliseconds**)	U – Q0.32	4294967123	0.9999999597202986

3.11 DCS – fast (deprecated)

Parameter group: 11

Parameter group: 7

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
~~Threshold_high~~	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**
~~Htreshold_low~~	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**
~~Enable_shift~~	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**	**DEPRECATED**

3.12 Peak detector – fast

Parameter group: 12

Setting	**Description**	**Signedness and fixed-point format (Qm.n)**	**Default value (unsigned 32-bit)**	**Default value (human readable)**
Tclk	Period of the DPP clock in the FPGA, in **nanoseconds** (**ns**).	U – Q32.0	20	20
Blanking_time	Period to disable peak detector after an event detection, expressed in **clock cycles (Tclk units)**. Requires b*lanking time factor* to be computed.	U – Q32.0	135	135
Time_over_threshold	ToT for peak detector, in **milliseconds**. Requires Time Over Threshold factor. (tau_pk_top_fast + tot_factor*tau_pk_fast)/Tclk	U – Q2.14	65	0.00396728515625
x_min	Low-level discriminator (LLD), in **Volts**.	S – Q2.14	65	0.00396728515625
x_max	High-level discriminator (HLD), in **Volts**.	S – Q2.14	32061	1.95684814453125
flags	Enable peak detector subsystem (bit 0)	U – Q32.0	1	1

3.13 Pileup rejector

Parameter group: 13

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
Preset counter	Preset timer for PuR, expressed in clock cyles (**Tclk units**). Preset = (guard_time_factor*shaper_slow_taupk+shaper_slow_taupk_top)/Tclk	U – Q32.0	255	255
Enable	Enable flag (bit 0)	U – Q32.0	1	1
Delay	Synchronization delay for PuR in clock cycles (**Tclk units**). Delay = (2*shaper_fast_taupk+shaper_fast_taup_top)/Tclk	U – Q32.0	20	20

3.14 High Voltage (only for PMT version)

Parameter group: 14

Setting	Description	Signedness and fixed-point format (Qm.n)	Default value (unsigned 32-bit)	Default value (human readable)
Set HV	Measured with 16-bit DAC. Use the following expression, being **x** the desired value in **Volts**. HV = $((x*2^{16})/1250+0.5)$	U – Q32.0	44564	850 Volts
Read HV (read-only)	Measured with 12-bit ADC. The HV in Volts can be determined with the following equation, being **x** de value read from the ADC. HV = $1250*x/2^{12}$	U – Q32.0	Read-only Example: 2785	Read-only Example measurement: 850 Volts

3.15 Variable-gain analog amplifier (VGA)

Parameter group: 15

Two board versions available, each featuring different DAC models to drive the VGA input. In board Version A, the AD5693 DAC is present and only fine gain tuning is available. Version B uses the AD5697R DAC, enabling both fine gain and coarse gain.

Values for version A

Setting	**Description**	**Signedness and fixed-point format (Qm.n)**	**Default value (unsigned 32-bit)**	**Default value (human readable)**
Fine gain Range: [1.0, 2.0]	Analog amplification before ADC. Fine gain, being **x** the requested value and **y** the corresponding DAC code to be sent. $y = \mathbf{x} * 2^{16}/6.6 + 0.5$	U – Q15.0	9930	1.0
Coarse gain	Fixed. No coarse gain available.	U – Q1.0	1	1.0

Values for version B

Setting	**Description**	**Signedness and fixed-point format (Qm.n)**	**Default value (unsigned 32-bit)**	**Default value (human readable)**
Fine gain Range: [1.0, 2.0]	Analog amplification before ADC. Fine gain, being **x** the requested value and **y** the corresponding DAC code to be sent. $\mathbf{y} = \mathbf{x} * 2^{12}/5.0 + 0.5$	U – Q11.0	820	1.0
Coarse gain Range: [1.0, 21.0]	Analog amplification before ADC. Coarse gain, being **x** the requested value and **y** the corresponding DAC code to be sent. $\mathbf{y} = ((0.6 * 2^{12})/2.5) * \log_{10}(\mathbf{x})$ $\mathbf{y} = 983.04 * \log_{10}(\mathbf{x})$	U – Q11.0	687	5.0

4 Documentation history

This document has been built upon the available source code files and other available documentation. It is intended to be used as a guide to interact with the MCA either through a command line or to build an API based on it.

Document Version	**Description**	**Author**	**Date**
1.0	Initial release. Validated with hardware through a serial command line interface.	I. Morales	2026/05/29